

4 Find the Laplace transform of $f(t) = A[e^{-\alpha t} - 2e^{-\gamma t}]u(t)$. Locate the poles and zeros of $F(s)$.

4 Find the Laplace transform of $f(t) = A[\cos(\beta t) - \sin(\beta t)]u(t)$. Locate the poles and zeros of $F(s)$.

9-7 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams:

(a) $f_1(t) = [5e^{-5t} - 10e^{-20t}]u(t)$

(b) $f_2(t) = [5\cos(10t) + 7\cos(20t)]u(t)$

9-9 Find the Laplace transform of the following waveform. Locate the poles and zeros of $F(s)$.

(a) $f_1(t) = \delta(t) - (625te^{-50t})u(t)$

(b) $f_2(t) = [5 + e^{-20t} - 6\cos(10t) + 2\sin(10t)]u(t)$

9-17 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{2(s+5)}{s(s+10)}$

(b) $F_2(s) = \frac{s^2}{(s+5)(s+10)}$

9-18 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{20(s+20)}{(s+10)^2 + 400}$

(b) $F_2(s) = \frac{20(s-20)}{(s+10)^2 + 400}$

9-24 Find the inverse transforms for the following functions:

$$(a) F_1(s) = \frac{30(s+2)}{s(s^2+4s+5)}$$

$$(b) F_2(s) = \frac{2s}{(s+4)(s^2+4s+8)}$$

9-32 Use the Laplace transformation to find the $y(t)$ that satisfies the following first-order differential equation:

$$\frac{dy}{dt} + 500y = [2500e^{-250t}]u(t) \quad \text{with} \quad y(0^-) = 0$$

9-45 Use the initial and final value properties to find the initial and final values of the waveform corresponding to the transforms in Problem 9-22. If either property is not applicable, explain why.

9-22 Find the inverse transforms of the following functions:

$$(a) F_1(s) = \frac{16}{(s+2)(s^2+12s+36)}$$

$$(b) F_2(s) = \frac{2(s^2+2)}{s(s^2+4)}$$

9-47 Use the initial and final value properties to find the initial and final values of the waveform corresponding to the following transforms. If either property is not applicable, explain why.

$$(a) F_1(s) = \frac{2(s^2+5s+6)}{(s+2)(s+6)(s+12)}$$

$$(b) F_2(s) = \frac{10(s^2+10s+40)}{s(s^2-100)}$$